

4.9 Chi-Squared Distribution

An important question in conducting scientific research is the question of **representativeness**. E.g. is a specific sample representative in terms of gender? In answering this question, use is made of the **chi-squared distribution**.

Definition

Suppose the variables $X_1, X_2, \dots, X_n$ follow a standard normal distribution and are mutually independent. The random variable

$$\chi^2 = X_1^2 + X_2^2 + \dots + X_n^2$$

is a **chi-squared variable with n degrees of freedom**.

Remarks

1. Notation: $\chi^2 \sim \chi^2(n)$

2. Since

$$\begin{aligned} V(X_1) &= E(X_1^2) - E(X_1)^2 && \Leftrightarrow && 1 = E(X_1^2) - 0^2 \\ &&& \Leftrightarrow && E(X_1^2) = 1 \end{aligned}$$

it follows that

$$\begin{aligned} E(\chi^2) &= E(X_1^2 + X_2^2 + \dots + X_n^2) \\ &= E(X_1^2) + E(X_2^2) + \dots + E(X_n^2) \\ &= 1 + 1 + \dots + 1 \\ &= n \end{aligned}$$

In a similar way

$$V(\chi^2) = 2n$$

In summary

$$\begin{aligned} E(\chi^2) &= n \\ V(\chi^2) &= 2n \end{aligned}$$

3. The shape of the graph of the probability density function of the **chi-squared distribution** χ^2 depends on the number of degrees of freedom. The higher the number of degrees of freedom, the more the graph is similar to that of the **normal distribution**.

So

$$\chi^2 \approx N(\mu = n; \sigma = \sqrt{2n}), \text{ provided } n \text{ sufficiently large.}$$

This result is based on the **Central Limit Theorem**. See Figure 9.3.

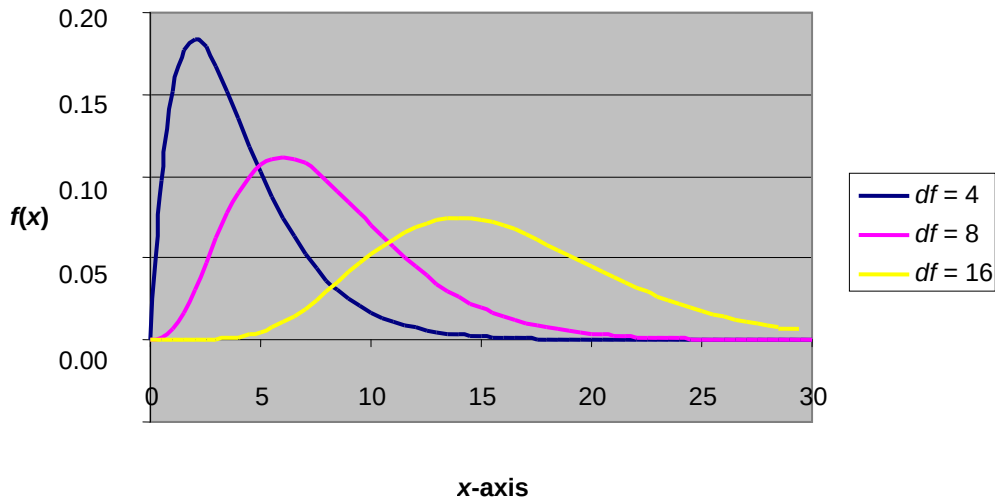


Figure 9.3 Chi-squared probability density functions for selected degrees of freedom

4. **Table IV** provides percentage points of the chi-squared distribution. It contains values of the random variable χ^2 above which we find a probability α . It is also possible to use the **graphical calculator**. In that case we choose the option $\chi^2\text{cdf}(a, b, df)$ (i.e. **gc/distr/8**).

Examples

- | | | | |
|----|-------------|---|-----------------|
| 4. | $df = 5:$ | $P(\chi^2 \geq 15.09) = 0.01$ | Table IV |
| | | $\chi^2\text{cdf}(a = 15.09, b = 10^{10}, df = 5) = 0.0100$ | gc |
| 5. | $df = 15:$ | $P(\chi^2 \leq 25.00) = 0.95$ | Table IV |
| | | $\chi^2\text{cdf}(-10^{10}, 25.00, 15) = 0.9501$ | gc |
| 6. | $df = 100:$ | $P(\chi^2 \geq 124.34) = 0.05$ | Table IV |

Remarks

- The number 124.34 in Example 6 is denoted by **g0.05; 100**.
- In the case of a random variable $X \sim N(\mu = df = 100; \sigma = \sqrt{2df} = \sqrt{200})$:

$$P(X \geq 124.34) = \text{normalcdf}(124.34, 10^{10}, 100, \sqrt{200}) = 0.0426$$

So an accurate **approximation** of the **chi-squared distribution** with the **normal distribution** only applies to large values of n .

9.7 Testing for Goodness-of-Fit

The **goodness-of-fit test** answers the question whether there is a certain **pattern** e.g. “Does the number of defects found in 60 identical products follow a **Poisson distribution**?”.

Number of defects	0	1	2	3	Total
Observed	32	15	9	4	60

Observed is given. **Expected** is calculated based on the assumption that there is a Poisson distribution.

If the differences between **Observed** (O_i) and **Expected** (E_i) are small, it is likely that there is a Poisson distribution.

We therefore determine:

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

$\left(\frac{O_i - E_i}{\sqrt{E_i}} \right)^2$

It applies: $\chi^2 \approx \chi^2(df = k - p - 1)$, with p = the number of estimated parameters (in this case: 1), provided that $E_i \geq 5$ ($\forall i$).

The only parameter to be estimated is λ .

Since

$$(32 \cdot 0 + 15 \cdot 1 + 9 \cdot 2 + 4 \cdot 3) / 60 = 0.75$$

λ is estimated at 0.75.

So

$$\begin{aligned} E_1 &= \text{Poissonpdf}(.75, 0) \cdot 60 = 28.34; & E_2 &= \text{Poissonpdf}(.75, 1) \cdot 60 = 21.26; \\ E_3 &= \text{Poissonpdf}(.75, 2) \cdot 60 = 7.97; & E_4 &= (1 - \text{Poissoncdf}(.75, 2)) \cdot 60 = 2.43. \end{aligned}$$

Since $E_4 < 5$, the classes 3 and 4 are combined. So $E_3 = (1 - \text{Poissoncdf}(.75, 1)) \cdot 60 = 10.40$.

The **test procedure** is now carried out as follows:

1. H_0 : $N = \text{number of defects} \sim \text{Poi}(\lambda = 0.75)$
 H_1 : $N \not\sim \text{Poi}(\lambda = 0.75)$
2. $\alpha = 0.05$
3. Test statistic:

$$\begin{aligned} \chi^2 &= \sum_{i=1}^3 \frac{(O_i - E_i)^2}{E_i} \sim \chi^2(df = 3 - 1 - 1 = 1), \text{ with:} \\ \chi^2 &= (32 - 28.34)^2 / 28.34 + (15 - 21.26)^2 / 21.26 + (13 - 10.40)^2 / 10.40 \approx 2.97 \end{aligned}$$

4. Critical region

$$P(H_1|H_0) \leq \alpha \quad \Leftrightarrow \quad P(\chi^2 > C|H_0) \leq 0.05$$

$$\Leftrightarrow \quad C = 3.84$$

Table IV

So $Z = C = g_{0.05; 1} = 3.84 ; \rightarrow$

5. Conclusion

Since $\chi^2 = 2.97 \notin Z$, we do not reject H_0 . So there is a Poisson distribution.

Remark

Statistical software packages (e.g. [Excel](#)) determine the test result by calculating the **P-value**. In that case:

$$\begin{aligned} \text{P-value} \leq \alpha & \Rightarrow \text{reject } H_0 \\ \text{P-value} > \alpha & \Rightarrow \text{do not reject } H_0 \end{aligned}$$

In the case of our example:

$$\begin{aligned} \text{P-value} &= P(\chi^2 \geq 2.97 | H_0) \\ &= \chi^2 \text{cdf}(2.97, 10^{10}, 1) \\ &= 0.0848 > \alpha = 0.05 \end{aligned}$$

Example 7

Given are 25 [interarrivaltimes](#). After arrangement from small to large:

0.01 0.03 0.04 0.04 0.05 0.07 0.08 0.10 0.10 0.11 0.11 0.19 0.22
0.27 0.29 0.36 0.46 0.51 0.54 0.56 0.72 0.73 0.93 1.17 1.50

Are these measurements taken from an [exponential distribution](#)?

We first estimate λ .

Since $\bar{t} = \frac{\sum t_i}{n} = \frac{9.19}{25}$, it follows that $\hat{\lambda} = \frac{25}{9.19} \approx 2.72$.

We use the following commonly used method to determine the **O-E-table**: choose k equally large classes such that $E_i \geq 5 (\forall i)$. So $k = 5$.

Class	Observed	Expected
< 0.082	7	5
[0.082; 0.188>	4	5
[0.188; 0.336>	4	5
[0.336; 0.592>	5	5
≥ 0.592	5	5

The class boundaries are determined by means of $F(t) = 1 - e^{-\lambda t}$:

$$F(t) = 0.2 \quad \Leftrightarrow \quad e^{-2.72t} = 0.8 \quad \Leftrightarrow \quad t = 0.082$$

The test procedure is carried out as follows:

1. $H_0: T \sim \exp(\lambda = 2.72)$
 $H_1: T \not\sim \exp(\lambda = 2.72)$

2. $\alpha = 0.05$

3. Test statistic number of estimated parameters

$$\chi^2 = \sum_{i=1}^5 \frac{(O_i - E_i)^2}{E_i} \sim \chi^2(df = 5 - 1 - 1 = 3), \text{ with:}$$

$$\chi^2 = (7 - 5)^2/5 + 2 \cdot (4 - 5)^2/5 + 2 \cdot 0 = 1.2$$

4. Critical region

$$P(H_1 | H_0) \leq \alpha \quad \Leftrightarrow \quad P(\chi^2 > C | H_0) \leq 0.05$$

$$\Leftrightarrow \quad C = 7.81$$

Table IV

So $Z = < C = g_{0.05, 3} = 7.81 ; \rightarrow >$

5. Conclusion

Since $\chi^2 = 1.2 \notin Z$, we do not reject H_0 .

So it is likely that the measurements are taken from an exponential distribution.

Remarks

1. $P\text{-value} = P(\chi^2 \geq 1.2 | H_0)$
 $= \chi^2 \text{cdf}(1.2, 10^{10}, 1)$
 $= 0.7530 > \alpha = 0.05$

2. Graphical calculator:

STAT $\rightarrow$ EDIT $\rightarrow$ Edit: $L_1 = \{7, 4, 4, 5, 5\}$, $L_2 = \{5, 5, 5, 5, 5\}$.

STAT $\rightarrow$ TESTS $\rightarrow \chi^2 \text{GOF-Test}$ with "Observed" = L_1 , "Expected" = L_2 and $df = 3$.

3. The requirement $E_i \geq 5 (\forall i)$ is in practice often relaxed to:
 - a. $E_i \geq 1 (\forall i)$;
 - b. $1 \leq E_i < 5$ in a maximum of 20% of the classes.